

Mohammad Sedky

Senior Java Backend Developer | Spring Boot · Microservices · Docker

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Professional Summary

Senior Java Backend Developer with 12+ years of experience designing and building scalable enterprise systems across Telecom, Banking, Microfinance, and Government domains. Skilled in **Spring Boot microservices** and relational databases (**PostgreSQL, Oracle, MySQL**) with exposure to **MongoDB**.

Experienced in **Dockerized deployments, Event-Driven Architecture (RabbitMQ), CI/CD pipelines**, and **system performance optimization**.

Currently working remotely with **STC (Saudi Telecom Company)** on large-scale property management platforms. Passionate about **distributed systems** and continuous improvement of backend performance and reliability.

Key Skills

Languages & Frameworks: Java (SE, J2EE), Spring Boot, Spring Cloud, JPA/Hibernate, Spring MVC, JSP, Oracle ADF, JSF, JavaScript, MapStruct

Databases: PostgreSQL, Oracle, MySQL, MongoDB, PL/SQL

DevOps & Tools: Docker, Jenkins, Kubernetes, Maven, Git, Linux, AWS, Redis, RabbitMQ (EDA, Outbox Pattern, Saga Choreography, Retry, DLQ, Manual ACK, Idempotency)

Testing & Reporting: JUnit, Mockito, Testcontainers, Jasper Reports

Professional Experience

Saudi Telecom Company (STC) — Riyadh, KSA (Remote via Digi Visions)

Senior Java Backend Developer | Sep 2023 – Present

- Developed and maintained backend microservices for **STC Square** (stcsquare.com.sa), a property rental and management platform proposed to **Aqalat KSA**, powering both the public website and internal admin dashboard.
- Designed and optimized REST APIs using **Spring Boot** and **PostgreSQL**, deployed via **Docker** and **Jenkins**.
- Additionally worked with **GraphQL** in internal systems such as **Royal Inventory** and **AMS** for flexible data querying.
- Improved dashboard performance by consolidating multiple API calls and optimizing SQL queries.
- Collaborated with QA, frontend, and business analysts to ensure smooth sprint deliveries.

- Contributed to production deployment, performance tuning, and debugging across multiple environments.

Other Projects within STC:

- **MMS (Meeting Management System):** Enhanced recurring meeting scheduling, reminder emails, and participant notifications.
 - **AMS (Asset Management System):** Co-developed from the ground up; built backend for asset allocation and ERP integration.
 - **Royal Inventory:** Managed workflows for converting request items into master inventory items.
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Egabi FSI — Cairo, Egypt

Senior Backend Java Developer | Mar 2022 – Aug 2023

- Led backend development for **Banking** and **Microfinance** systems serving **CIB, Credit Agricole, Bedayti, and Reefy**.
 - Owned and maintained **I-Score integration modules** across all microfinance platforms, ensuring compliance with **Central Bank of Egypt** standards and automating inquiry workflows.
 - Developed modules for **MBRHE (Dubai)** housing loans and the **Egyptian Microfinance Federation (EMFF)** for managing loan lifecycles.
 - Implemented reporting services using **Jasper Reports** for generating financial and performance data.
 - Mentored and supervised junior developers, providing technical guidance and code reviews.
 - Collaborated with QA and client teams to support UAT and resolve production issues.
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Golf Veterinary Center — Cairo, Egypt

Senior Java Developer | Oct 2020 – Feb 2022

- Designed and implemented a **Veterinary Management System** covering booking, accounting, and reporting modules.
 - Developed a **monolithic Spring MVC and JSP-based application**, later initiating migration of the booking service into a **Spring Boot microservice**.
 - Deployed and maintained the application on cloud hosting and provided production support.
 - Built **reporting** and **dashboard modules** for operations and accounting
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Earlier Experience

Data Management Systems (DMS) — Java Web Developer | Dec 2018 – Oct 2020

- Developed hospital management modules (registration, outpatient, inpatient) using **JSF**, **Oracle**, and backend Java.
- Built dynamic UI components using **JavaScript** and **jQuery**.
- Integrated frontend views with backend services and optimized user workflows.
- Improved database performance using optimized SQL queries.

Modern Telecom Systems (MTS) — Oracle ADF Developer | *Jun 2016 – Apr 2018*

Developed and maintained governmental applications for Egyptian ministries (Health, Transport, Telecom Egypt). Implemented PL/SQL procedures and business workflows.

Ultimate Soft / Pylon Systems / New Way — Java Developer | *Jun 2015 – Apr 2016*

Worked on encryption systems, accounting platforms, and parental-control applications using Java and Oracle.

Personal Projects

Event-Driven Microservices with RabbitMQ

- Built asynchronous communication using RabbitMQ between microservices
- Demonstrated event-driven architecture and eventual consistency
- Used MySQL and MongoDB (polyglot persistence)

GitHub: <https://github.com/msedky/rabbitmq-manual-ack-sample>

RabbitMQ Retry & Dead Letter Queue (DLQ) Handling

- Implemented retry and DLQ handling for fault-tolerant messaging
- Differentiated between retryable and non-retryable failures
- Performed manual recovery from DLQ (inspect → fix → replay → cleanup)

GitHub: <https://github.com/msedky/rabbitmq-retry-dlq-sample>

Idempotent Payment Processing with RabbitMQ

- Implemented idempotent payment processing using Idempotency-Key and request hashing to prevent duplicate transactions
- Designed handling for retry, duplicate, concurrent, and conflicting requests with proper response caching and 409 conflict detection
- Ensured safe interaction with external PSP under timeout and retry scenarios without creating duplicate payments
- Integrated idempotency with event-driven architecture using RabbitMQ to guarantee reliable and consistent processing

GitHub: <https://github.com/msedky/rabbitmq-idempotent-payment-case-study>

RabbitMQ Outbox Pattern — Order Fulfillment Case Study

- Guaranteed atomic DB writes and event publishing across 3 microservices using the Outbox Pattern
 - Implemented full event-driven order lifecycle with Saga Choreography compensation flows
 - Applied DLQ handling, MANUAL ACK, retry classification, and Strategy Pattern for event publishing
 - Deployed on Kubernetes with Secrets, ConfigMaps, and Ingress — covered with unit and integration tests
- GitHub: <https://github.com/msedky/rabbitmq-outbox-order-fulfillment>

JVM Memory Leak Case Studies — Spring Boot

- Reproduced, diagnosed, and fixed 3 real-world OutOfMemoryError scenarios in Spring Boot
 - Diagnosed each case using Heap Dump analysis with Eclipse MAT (Leak Suspects, Dominator Tree, Reference Chains)
 - Scenarios: Unbounded Cache, Unregistered Event Listeners, Large Result Set without Pagination
 - Load tested with k6 and analyzed JVM behavior under memory pressure
- GitHub: <https://github.com/msedky/springboot-memory-leak-case-study>

Education

MSc Computer Science, Arab Academy for Science, Tech & Maritime Transport
– 2015

BSc Computer Science, Arab Academy for Science, Tech & Maritime Transport
– 2010

GPA: 3.54 (Excellent)

Certifications

- Oracle Certified Professional, Java SE 8 Programmer
- Oracle PL/SQL Developer Certified Associate

Academic Highlights

Published two peer-reviewed papers on secure e-government and cryptographic algorithms, presented at the **Science & Information Conference (London, 2015)** and published in the **European Journal of Science (2014)**.

Languages: Arabic (Native), English (Very Good)

Nationality: Egyptian